

Simulation Without a Black Box

The Clinic Case Study

What We Have

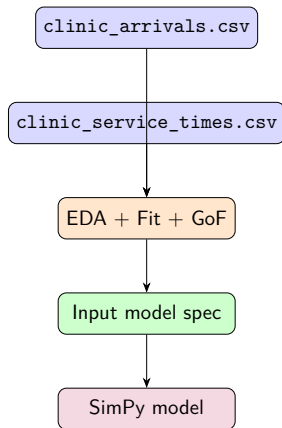
Two CSV files collected over 8 weeks at a primary-care clinic:

- `clinic_arrivals.csv` — 847 interarrival times (minutes)
- `clinic_service_times.csv` — 847 service durations (minutes)

Goal: Fit probability distributions to both files so that the SimPy clinic model uses realistic stochastic inputs rather than fixed constants. **Process stages modeled:**

1. Patient interarrival time $\rightarrow$ drives arrival process
2. Consultation service time $\rightarrow$ drives server utilization

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EDA Summary for Both Datasets

Dataset	n	Mean	Std	CV	Skewness	Min / Max
Interarrivals	847	4.23	4.18	0.99	1.12	0.03 / 31.4
Service times	847	12.7	7.8	0.61	0.87	1.2 / 58.3

Interarrival interpretation:

- $CV \approx 1.0 \Rightarrow$ Exponential is the natural candidate
- Skewness > 1 supports a right-skewed, memoryless process
- Poisson arrivals at rate $\lambda = 1/4.23 \approx 14.2/\text{hr}$

Service time interpretation:

- $CV \approx 0.61 \Rightarrow$ Gamma or Erlang, *not* Exponential
- Using Exponential would **overestimate** W_q by 30–50%
- Max value 58.3 min suggests occasional complex consultations

AIC Comparison: Which Family Wins?

Interarrival times:

Family	AIC	Δ AIC
Exponential	4821	0
Gamma ($\alpha = 1.08$)	4823	2
Weibull	4828	7
Lognormal	4851	30

Exponential wins; Gamma is essentially tied (Δ AIC = 2).

Service times:

Family	AIC	Δ AIC
Gamma ($\alpha = 2.65$)	5312	0
Lognormal	5319	7
Weibull	5327	15
Exponential	5398	86

Selected Input Model

- Interarrivals:** Exp($\mu = 4.23$)
- Service times:** Gamma($\alpha = 2.65$, $\beta = 4.79$)

```
from scipy import stats
import numpy as np
```

```
ia = pd.read_csv(
    "clinic_arrivals.csv")["ia_min"]
svc = pd.read_csv(
    "clinic_service_times.csv")["svc_min"]
```

```
# Interarrivals
_, sc_ia = stats.expon.fit(ia, floc=0)
```

```
# Service times
a_s, _, sc_s = stats.gamma.fit(
    svc, floc=0)
```

KS and Chi-Squared Test Results

Dataset	Model	KS D_n	KS p	χ^2	df	$\chi^2 p$
Interarrivals	Exponential	0.028	0.52	9.4	8	0.31
Service times	Gamma	0.031	0.39	11.2	7	0.13
Service times	Lognormal	0.048	0.04	18.7	7	0.01

Conclusions:

- Exponential fits interarrivals well
($p = 0.52$, no rejection)
- Gamma fits service times well
($p = 0.39$, $p = 0.13$)
- Lognormal *marginally rejected* by KS
($p = 0.04$) and chi-squared ($p = 0.01$)
— consistent with Q-Q upper-tail bow

Warning

With $n = 847$, a small KS statistic (0.048) that triggers rejection may be operationally negligible. Always pair statistical tests with Q-Q inspection.

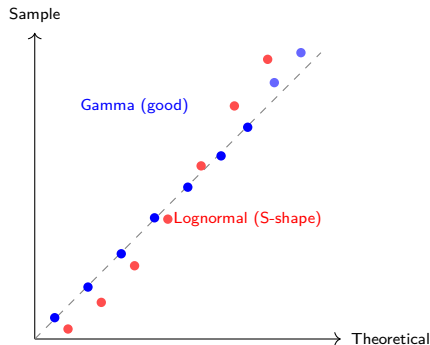
Q-Q Plots: What the Pictures Tell Us

Gamma Q-Q (service times):

- Points hug the 45° line from 0 to ~35 min
- A few points above the line above 40 min: minor heavy-tail deviation, likely complex consultations
- Verdict: **adequate** for simulation

Lognormal Q-Q (service times):

- S-shape in the body — Lognormal assigns too much probability to very short times
- Upper tail better, but overall worse than Gamma
- Verdict: **reject** in favor of Gamma



Professional Input Model Specification

Four-Sentence Format

One sentence each for: (1) the process, (2) the data source, (3) the chosen distribution and parameters, (4) the evidence.

Interarrival model:

Patients arrive at the clinic according to an arrival process whose successive interarrival times were measured over 8 weeks. Data source: electronic check-in logs, $n = 847$ records, verified against appointment schedules. We model interarrival times as $\text{Exp}(\mu = 4.23 \text{ min})$, corresponding to $\lambda \approx 14.2$ patients/hour. Evidence: KS $D_{847} = 0.028$ ($p = 0.52$); Q-Q plot shows no systematic deviation.

Service time model:

Consultation durations at the physician's room were recorded for each of the 847 visits. Data source: EHR timestamps, filtered to remove incomplete records. We model service times as $\text{Gamma}(\alpha = 2.65, \beta = 4.79 \text{ min})$, giving $\mu = 12.7 \text{ min}$ and $CV = 0.61$. Evidence: KS $D_{847} = 0.031$ ($p = 0.39$); AIC 86 units below Exponential.

Sensitivity: What If We Used Lognormal?

Replace $\text{Gamma}(\alpha = 2.65, \beta = 4.79)$ with $\text{Lognormal}(\mu_{\ln} = 2.35, \sigma_{\ln} = 0.61)$ (same mean, same CV). Run 50 replications of the clinic model; compare $\bar{W}_q$.

Input model	$\bar{W}_q$ (min)	95% CI
Gamma (selected)	18.4	[17.1, 19.7]
Lognormal (alt.)	22.8	[20.9, 24.7]
Difference	4.4	[2.1, 6.7]

Implication: Using the wrong family inflates predicted wait time by $\sim 24\%$, which could lead to over-staffing decisions.

Warning

“Close enough” in AIC does not mean “equivalent for decisions.” A ΔAIC of 7 between Gamma and Lognormal translates to a 24% error in W_q near $\rho = 0.85$.

NORTA Preview

When arrival and service times are *correlated* (busy periods have longer consultations), use the **NORmal-To-Anything (NORTA)** method to generate correlated marginals. Covered in Module M10.

Key Takeaways

1. **EDA first:** CV of 0.99 (interarrivals) versus 0.61 (service) tells you to use different families for different processes.
2. **AIC gap matters:** Exponential loses to Gamma for service times by $\Delta\text{AIC} = 86$ — not a close call.
3. **GoF and Q-Q together:** Lognormal fails KS and shows S-shape; Gamma passes both.
4. **Document with four sentences:** process, source, model, evidence.
5. **Sensitivity test:** swapping to the second-best distribution changed W_q by 24% — always check.
6. **Correlated inputs** require NORTA; independence assumption is not always safe.

Next Module: M08 — Output Analysis

L26 — Initialization bias and Welch's warmup procedure.